

PROBLEM SET 3: ENDOGENOUS INNOVATION

PROBLEM I – OPTIMALITY IN THE R&D BASED ENDOGENOUS GROWTH MODEL

Consider the following version of the Grossman-Helpman model:

$$\begin{aligned} U_t &= \int_t^{+\infty} (\ln C_t) e^{-\rho(s-t)} ds \\ C_t &= \left(\int_0^{N_t} c_{it}^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}} \\ p_{it} &= \mu w_t \\ \frac{\dot{N}_t}{N_t} &= L_{Rt} \\ c_{it} &= l_{it} \\ \int_0^{N_t} l_{it} di &= L_{Pt} = L - L_{Rt} \end{aligned}$$

That is, we assume $\gamma = 1$, a logarithmic utility, and the monopoly markup is now exogenous.

1 – Compute the equilibrium growth rate and discuss how it is affected by the monopoly markup

2 – Assume now that L_{Rt} is chosen by a central planner who maximizes the consumer's utility $\int_t^{+\infty} (\ln C_t) e^{-\rho(s-t)} ds$ subject to the constraints

$$C_t = N^{\frac{1}{\sigma-1}} (L - L_{Rt})$$

and

$$\frac{\dot{N}_t}{N_t} = L_{Rt}$$

Write the Hamiltonian and the first order conditions. Compute the optimal growth rate of the economy.

3 – What is the monopoly markup which deliver the optimal growth rate as an equilibrium? What happens if the markup is larger than this level? Smaller?

PROBLEM II – A SEMI-ENDOGENOUS GROWTH MODEL

Consider the following model.

Total population grows at rate n . At date t it is equal to L_t .

While population grows, we assume that a representative "dynasty" determines the consumption path. Its utility function is

$$\int_0^{+\infty} \ln c_t \cdot e^{-\rho t} dt,$$

where c_t is the per-capita aggregate consumption index at t . Denoting total consumption by C_t , we have $C_t = c_t L_t$.

Let N_t be the mass of goods at date t . Then the aggregate consumption index is

$$c_t = \left(\int_0^{N_t} c_{it}^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}},$$

where c_{it} is per-capita consumption of good i .

At each date t a fraction β_t of the population does R and D. The production function for new goods is

$$\dot{N}_t = \gamma N_t^\delta \beta_t L_t.$$

The term in N_t^δ is a productive externality from the stock of knowledge to the productivity in generating new knowledge. We assume $\delta < 1$.

Each individual good is produced by a monopoly. The production function is

$$y_i = l_i,$$

where y_i is the output of good i and l_i is employment in good i .

Once a good is invented, the inventor owns a patent which gives him the monopoly rights over the production of that good forever.

We will try to compute a balanced growth path of this economy. We denote the real wage at t by w_t , the price of good i by p_{it} , the aggregate price level by p_t , and the real interest rate by r_t .

We will normalize the aggregate price level to $p_t = 1$.

1 – Write down the intertemporal budget constraint of the representative dynasty.

2 – Show that the Euler equation for consumption is

$$\frac{\dot{C}_t}{C_t} = r_t - \rho,$$

or equivalently

$$\frac{\dot{c}_t}{c_t} = r_t - n - \rho.$$

3 – What is the expression of the aggregate price level p_t as a function of the p_{it} 's?

4 – Derive a simple relationship between p_{it} and w_t .

5 – What is the wage at t as a function of N_t ?

6 – What is the quantity of any good being produced at t , given N_t and β_t ?

7 – What is the level of monopoly profits π_t ?

8 – Write down the evolution equation for the value of an individual firm V_t .

9 – Derive a relationship between V_t , N_t and w_t coming from the free entry condition in the R and D sector.

10 – Let g be the growth rate of the aggregate per capita consumption index, that is, $g = g_c$. Show that in a balanced growth path $g_N = (\sigma - 1)g$ and $g_w = g$.

11 – Show that

$$g = \frac{n}{(1 - \delta)(\sigma - 1)}.$$

12 – Explain the role played by n in the determination of g .

13 – Compute the value of the interest rate along the BGP.

14 – Do R&D incentives affect long-run growth in this model? If not, what effect do they have on the economy?